

IMMUNOBIOLOGY AND IMMUNOTHERAPY

Product-intrinsic NF- κ B-driven transcriptional programs connote durability of CAR-T response in multiple myeloma

Jerald D. Noble,^{1,*} Barbara C. Peixoto,^{1,*} Meghan A. Menges,¹ Julieta Abraham-Miranda,¹ Constanza Savid-Frontera,¹ William Sawyer,¹ Vasu D. Sorathia,¹ Luis A. Cuadrado Delgado,² Salvatore A. Corallo,² Julia C. Llanos,² Emily C. Merritt,² Gabe de Avila,² Omar A. Castaneda Puglianini,² Hien Liu,² Melissa Alsina,² Taiga Nishihori,² Kenneth H. Shain,³ Ariosto S. Silva,⁴ Rachid C. Baz,⁵ Brandon J. Blue,³ Ariel F. Grajales-Cruz,³ Doris K. Hansen,² John L. Cleveland,⁵ Fabiana Perna,⁴ Conor C. Lynch,⁵ Jennifer M. Binning,⁶ Reginald M. Atkins,¹ Xiaofei Song,⁷ Frederick L. Locke,^{2,†} and Ciara L. Freeman^{2,†}

¹Department of Clinical Science, ²Department of Blood and Marrow Transplant and Cellular Immunotherapy, ³Department of Malignant Hematology, ⁴Department of Metabolism and Physiology, ⁵Department of Tumor Microenvironment and Metastasis, ⁶Department of Molecular Oncology, and ⁷Department of Biostatistics and Bioinformatics, H. Lee Moffitt Cancer Center & Research Institute, Tampa, FL

KEY POINTS

- Ide-cel product NF- κ B activity is associated with longer progression-free survival in multiple myeloma.
- This signature marks a global T-cell fitness state present before CAR-T manufacture.

Idecabtagene vicleucel (ide-cel) induces deep responses in relapsed/refractory multiple myeloma, yet more than half of patients relapse within 1 year. The intrinsic features of chimeric antigen receptor T-cell (CAR-T) products that distinguish durable from nondurable responders are poorly defined, particularly at single-cell resolution, and defining drivers of durable response is critical to guide patient counseling and to inform strategies for optimizing CAR-T manufacturing and efficacy. To address this need, 40 ide-cel infusion products (184 398 cells) were profiled using single-cell RNA sequencing. These analyses revealed that a transcriptional program in CD4 CAR-T cells that led to durable responses is characterized by NF- κ B signaling, prosurvival circuits, tonic/chemokine signaling, and elevated CAR transgene expression. These features were associated with prolonged progression-free and overall survival irrespective of baseline clinical characteristics. Further, analyses of paired apheresis and tumor microenvironment samples

showed that elevated NF- κ B activity is an intrinsic hallmark of T-cell fitness that is characterized by a central memory phenotype and the lack of checkpoint receptor expression, and that these features were manifest in marrow-derived and peripheral blood T cells before CAR-T manufacturing. Finally, validating functional relevance, the pharmacologic inhibition of NF- κ B abrogated CAR-T cytotoxicity and cytokine production in vitro. Our results support that NF- κ B in the ide-cel product marks a signaling axis affecting CAR-T function and that NF- κ B activity represents a global marker of T-cell fitness present before CAR-T manufacture.

Introduction

Chimeric antigen receptor T-cell (CAR-T) therapy targeting B-cell maturation antigen (BCMA) has significantly advanced treatment options for patients with relapsed/refractory multiple myeloma (R/R MM), yielding deep responses even in heavily pretreated populations.^{1–6} Despite these successes, however, durable remissions are not observed for the majority of treated patients,⁷ and the drivers of durable responses after CAR-T cell infusion are poorly understood.

Although product-specific features such as T-cell phenotype,^{7–11} transduction efficiency,¹² and in vivo expansion^{7,13} have been

associated with clinical outcomes, the transcriptional states of preinfused T cells remain underexplored. Transcriptional programs governing activation, survival, and persistence of CAR-T cells in vivo^{14–17} are likely to influence CAR expression and therapeutic efficacy, yet it is unclear whether such programs are manifest in clinical-grade infusion products (IP) and forecast long-term benefit in R/R MM. To address this gap, we integrated phenotypic and gene expression data from matched pre-CAR-T apheresis and tumor microenvironment (TME) samples from a cohort of real-world patients treated with idecabtagene vicleucel (ide-cel). Notably, these analyses identified key features of T-cell fitness that are linked to therapeutic durability.

Methods

Patients with R/R MM treated with commercial ide-cel and who consented to a biospecimen protocol (Pro00021733) were included if IP was available (N = 40). Matched serum (when available) was used to measure soluble BCMA (sBCMA), as previously described.¹⁸ Apheresis was available for immunophenotyping in a subset (N = 20). Preinfusion bone marrow aspirates were available for paired analyses and for baseline TME profiling in an expanded refractory myeloma cohort (total N = 30; 9 with paired IP samples). Diagnostic marrow evaluation and fluorescence in situ hybridization testing were obtained as part of clinical care. Durable responders (DR) were defined as alive, progression-free, and requiring no additional therapy ≥ 9 months after infusion; nondurable responders (NDR) progressed or died before this time point. Progression-free survival (PFS) and overall survival (OS) were measured from infusion and compared using Kaplan-Meier/log-rank; Cox models were used for time-to-event associations.

Viable cryopreserved ide-cel IP was processed for single-cell RNA sequencing (scRNA-seq) on 10x Genomics Chromium Single Cell 5' version 2 and sequenced on an Illumina NovaSeq 6000. Reads were aligned with Cell Ranger version 7.1.0 to GRCh38 augmented with an ide-cel transgene sequence; single-nucleotide discrepancies were corrected using a published pipeline. CAR⁺ cells were defined as cells with ≥ 1 read mapping to the transgene.

After quality control and doublet removal, data were integrated in Seurat version 5 with Harmony. Populations of CD4 and CD8 cells were identified and analyzed separately (supplemental Figure 1, available on the *Blood* website). CD4 and CD8 cell populations were assessed for variance components (supplemental Figures 2-3), integrated using Harmony, and subjected to clustering. Uniform Manifold Approximation and Projection confirmed that clustering was not driven by individual samples, patient age, patient sex, sequencing batch, or CAR⁺ status (supplemental Figures 4-9). For each patient, CAR⁺ CD4 and CAR⁺ CD8 cells were pseudobulked, variance-stabilized in DESeq2, and scored with gene set variation analysis (gsva function) using prespecified CAR-T gene sets and a c-REL-dependent program. A convergence score was computed by z-scoring the antiapoptosis, tonic signaling, and c-REL knockout signature scores and averaging the 3 z-scores per patient.

Flow cytometry of IP, apheresis, and marrow used viability staining/Fc block; IP CAR-T enrichment used fluorescein isothiocyanate-conjugated BCMA protein with anti-fluorescein isothiocyanate magnetic microbeads (Miltenyi). Samples were acquired on a BD FACSymphony and analyzed in FlowJo. For baseline TME profiling, cryopreserved marrow was thawed, CD138-depleted (Miltenyi), RNA-extracted (Qiagen AllPrep), and profiled with the NanoString PanCancer Immune Profiling panel. For functional validation, healthy-donor peripheral blood mononuclear cells were activated with CD3/CD28 beads and transduced with an anti-BCMA CAR (4-1BB/CD3 ζ). NF- κ B was inhibited with the I κ B kinase β inhibitor BMS-345541 (7.5 μ M for 2 hours) followed by washout and confirmed by reduced phospho-p65. Cytotoxicity against BCMA⁺ NCI-H929 targets was assessed by a 24-hour luciferase killing assay using

NCI-H929 cells expressing green fluorescent protein (GFP) and luciferase (NCI-H929/GFP-Luc) with supernatants assayed for interferon gamma, interleukin-2 (IL-2), IL-6, and tumor necrosis factor α and by IncuCyte imaging using GFP-expressing targets. Repeated antigen exposure used serial stimulation on plate-bound recombinant human BCMA every 72 hours with inhibitor pretreatment; proliferation/viability and differentiation/exhaustion markers were measured by flow cytometry. Additional methods are described in depth in the supplemental Methods.

This retrospective study was approved by the Moffitt Cancer Center and Advarra institutional review boards and conducted per the Declaration of Helsinki.

Results

Patient characteristics

The baseline characteristics of 40 consecutive patients treated with ide-cel and having available IP are outlined in supplemental Table 1. This cohort is representative of real-world patients with a median age of 67 years (range, 43-81) and a median of 6 prior lines of therapy.² All patients were triple-class exposed in line with the approved indication; over half (63%) had undergone autologous stem cell transplantation (ASCT), and 38% were penta-refractory at the time of CAR-T infusion. High-risk cytogenetics were present in 37% of patients, most commonly 1q gain/amplification. A substantial proportion (58%) received bridging therapy, 15% had extramedullary disease before treatment, and 30% had $\geq 50\%$ plasma cell infiltration in the marrow before infusion. Median sBCMA levels were 93.4 ng/mL at apheresis and 119.4 ng/mL before lymphodepletion, again aligned with published data representing a real-world cohort.¹⁸ After infusion, residual IP was recovered via postinfusion bag elution, viable cryopreserved, and processed using a standardized thaw-wash-resuspension workflow immediately before single-cell encapsulation, minimizing pre-analytical variability and preserving product-intrinsic biology. Median duration of follow-up for living patients was 28.9 months (range, 7.1-44.1). All patients either had sufficient follow-up to exceed 9 months or experienced disease progression (NDR) (Figure 1A). At the 9-month landmark, 26 patients (65%) had DR and 14 (35%) had NDR.

Ide-cel IP cell subtypes have limited association with efficacy

After stringent quality control procedures (see "Methods"), a total of 184 398 single cells from CAR-T IP were analyzed. Individual cells were classified as CD4 or CD8 T cells using an established in silico gating strategy adapted from Li et al¹⁹ (supplemental Figure 1), yielding 108 818 (351-6081 per patient; interquartile range [IQR], 1901-3280) CD4 T cells and 41 203 (68-5423 per patient; IQR, 355.8-1322.8) CD8 T cells (Figure 1B-C). The populations of CD4 and CD8 cells were then independently integrated, clustered, and annotated based on canonical lineage and functional markers (Figure 1D-E; supplemental Figures 10-11), which identified 7 CD4 and 5 CD8 cell subtypes. To detect CAR-expressing cells within the scRNA-seq data, a customized reference transcriptome was constructed by incorporating the ide-cel CAR sequence into the GRCh38 human reference genome. Libraries were aligned

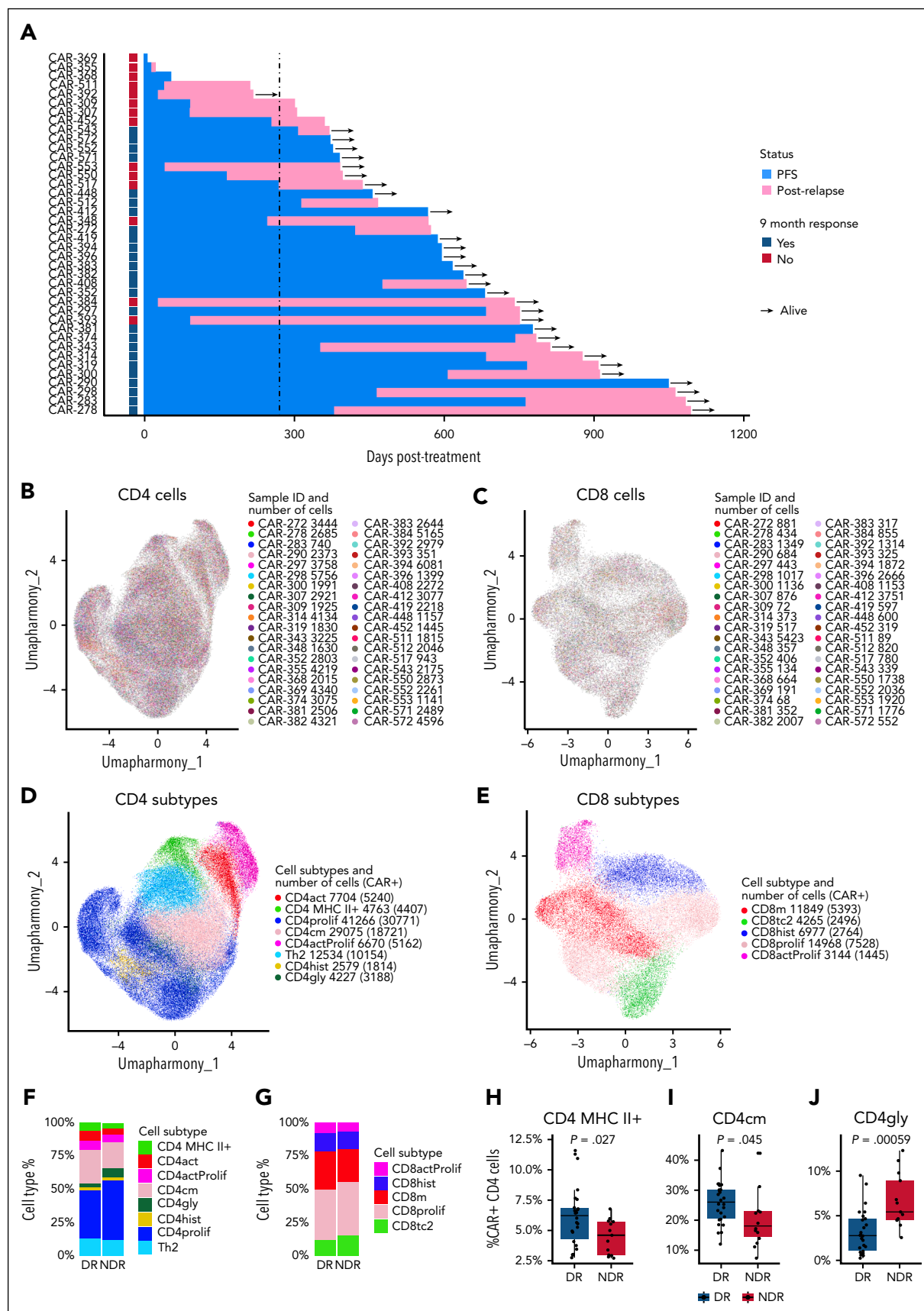


Figure 1. scRNA-seq of ide-cel IP identifies cell subtypes associated with durable response. (A) Swimmer plot depicting PFS and survival of the 40 patients with evaluable ide-cel product. (B-C) UMAPs depicting the number of cells per sample in CD4 (B) and CD8 (C) cell subsets. (D-E) CD4 (D) and CD8 (E) cell subtypes identified via

to this modified reference, and cells were designated as CAR⁺ if they contained at least 1 transcript mapping to the transgene (see “Methods”), enabling the detection of 79 457 (245-4631 cells per patient; IQR, 1370-2442) CAR⁺ CD4 T cells and 16 326 (31-2971 cells per patient; IQR, 190.2-659.5) CAR⁺ CD8 T cells (supplemental Figure 9).

Despite high-resolution profiling of IP, minimal cellular composition differences were observed between patients who experienced durable responses (DR) and those who did not (NDR) (Figure 1F-H; supplemental Figures 12 and 13). The most notable findings included greater CAR⁺ CD4 cells highly expressing MHC class II genes (Figure 1F; $P = .014$) and cells with an effector memory phenotype (Figure 1G; $P = .042$) in patients with durable responses and cells highly expressing glycolysis genes (Figure 1H; $P = .0003$) in patients with nondurable responses, features consistent with differentiated effector cells having high glycolytic and proliferative activity but limited metabolic flexibility and persistence required for sustained CAR-T function in vivo.

Efficacy is linked to prosurvival programs, including NF- κ B signaling and elevated CAR density in the product

Given the paucity of composition differences in IP cell subtypes between patients with durable responses and those with nondurable responses, we assessed if transcriptional signatures and functional states in CAR⁺ CD4 and CD8 cells were associated with durable response. Patients classified as DR had significantly higher proportions of CAR⁺ cells, that is, the proportion of IP cells that expressed at least 1 copy of ide-cel (Figure 2A). Moreover, among CAR⁺ cells, CD4 cells from patients classified as DR expressed higher levels of CAR transcripts per cell (Figure 2B), and this correlated with surface CAR expression as measured by flow cytometry (Figure 2C), suggesting CAR density may serve as a functional proxy for potency.

To compare gene signature differences at the sample level, we pseudobulk aggregated single-cell data from CAR⁺ CD4 and CD8 cells and evaluated 32 curated pathways from prior CAR-T literature via gsva within the gene set variation analysis²⁰ package to calculate patient-level gene signature scores (see “Methods”). This strategy provided coverage of T-cell-intrinsic biology across axes of activation, cytokine responses, differentiation, metabolism, proliferation, survival, stress resistance, and trafficking.

Clinical relevance was assessed by Cox models for PFS and OS and by DR vs NDR comparisons, with greater weight placed on PFS given its relevance to durable responses and reduced confounding from post-relapse therapies; false discovery rate-adjusted P values for these pathway-level analyses are provided in supplemental Table 2.

Across panels, multiple high-ranking signatures converged on shared biological themes, including NF- κ B signaling,

prosurvival/antiapoptotic programs, and 4-1BB-associated tonic signaling (supplemental Figure 14; supplemental Table 2). In CAR⁺ CD4 cells, these programs were associated with DR (Figures 2D-F), PFS (Figures 2G-I), and OS (Figures 2J-L). CD4 NF- κ B signaling score was not associated with peak postinfusion CAR copy number (Spearman $\rho = 0.032$; $P = .85$), suggesting that the favorable clinical association of this program is not reducible to greater peak CAR expansion alone (supplemental Figure 15). In CAR⁺ CD8 cells, the antiapoptosis signature was associated with PFS and OS (supplemental Figure 16), and tonic signaling was associated with DR (supplemental Figure 17).

To further interrogate the NF- κ B axis, we applied the c-REL-dependent signature defined by Lalle et al,²¹ derived from c-REL knockout experiments (higher scores indicate intact c-REL/NF- κ B activity). In ide-cel IP, higher scores for this signature in CAR⁺ CD4⁺ cells were associated with DR, and higher scores in CAR⁺ CD8 cells were associated with longer OS (supplemental Figure 18; supplemental Table 1). These findings mirror those of Lalle et al,²¹ who showed that c-REL-dependent programs marked responders and improved survival after PD-1 blockade in melanoma, bladder, and gastric cancers, supporting NF- κ B/c-REL activity as a product-intrinsic correlate of durable benefit.

Finally, there was a strong positive correlation between CAR transcript abundance and NF- κ B signaling, antiapoptosis, and tonic signaling gene signatures in CAR⁺ CD4 cells (Figure 2M-O), although this did not meet significance in CAR⁺ CD8 cells, except for tonic signaling (supplemental Figure 19A-C). These 3 pathways were intercorrelated (supplemental Figure 19D-E) in CAR⁺ CD4 cells, characterizing DR (Figure 2P). A convergence score calculated by averaging the z-scores of the antiapoptosis, tonic signaling, and c-REL knockout signatures for each patient (see “Methods”) revealed this convergence score is significantly correlated with NF- κ B signaling (supplemental Figure 20).

Pretreatment composition of circulating and tumor-resident T cells shapes NF- κ B pathway activation in the IP

Given the associations between CAR expression, tonic signaling, and antiapoptotic programs and the recognized convergence of these pathways on NF- κ B signaling, we assessed if NF- κ B serves as a central node that controls CAR-T cell persistence and function. Comparing baseline characteristics of patients categorized into high or low NF- κ B signature groups (Table 1) showed no significant differences between these groups regarding clinical prognostic factors, including the presence of high-risk cytogenetic abnormalities, sBCMA levels, inflammatory markers, bone marrow plasma cell burden, absolute lymphocyte count at the time of apheresis, CD4:CD8 T-cell ratios, the presence of extramedullary disease, or prior exposure to BCMA-directed therapy. Similarly, there were no differences in the proportion of patients with high vs low CAR-HEMATOX scores,²² suggesting differential clinical risk profiles do not account for observed disparities in PFS. There was a

Figure 1 (continued) clustering based upon scRNA-seq, the number of total cells of each subtype, and in parentheses, the number of CAR⁺ cells of each subtype. (F-G) Aggregated CAR⁺ CD4 (F) and CD8 (G) cell subtype compositions of DR and NDR. (H-J) Differences of cellular subtype composition between DR that were significant or that approached significance. P values were calculated using the unpaired Wilcoxon rank-sum test. UMAP, Uniform Manifold Approximation and Projection.

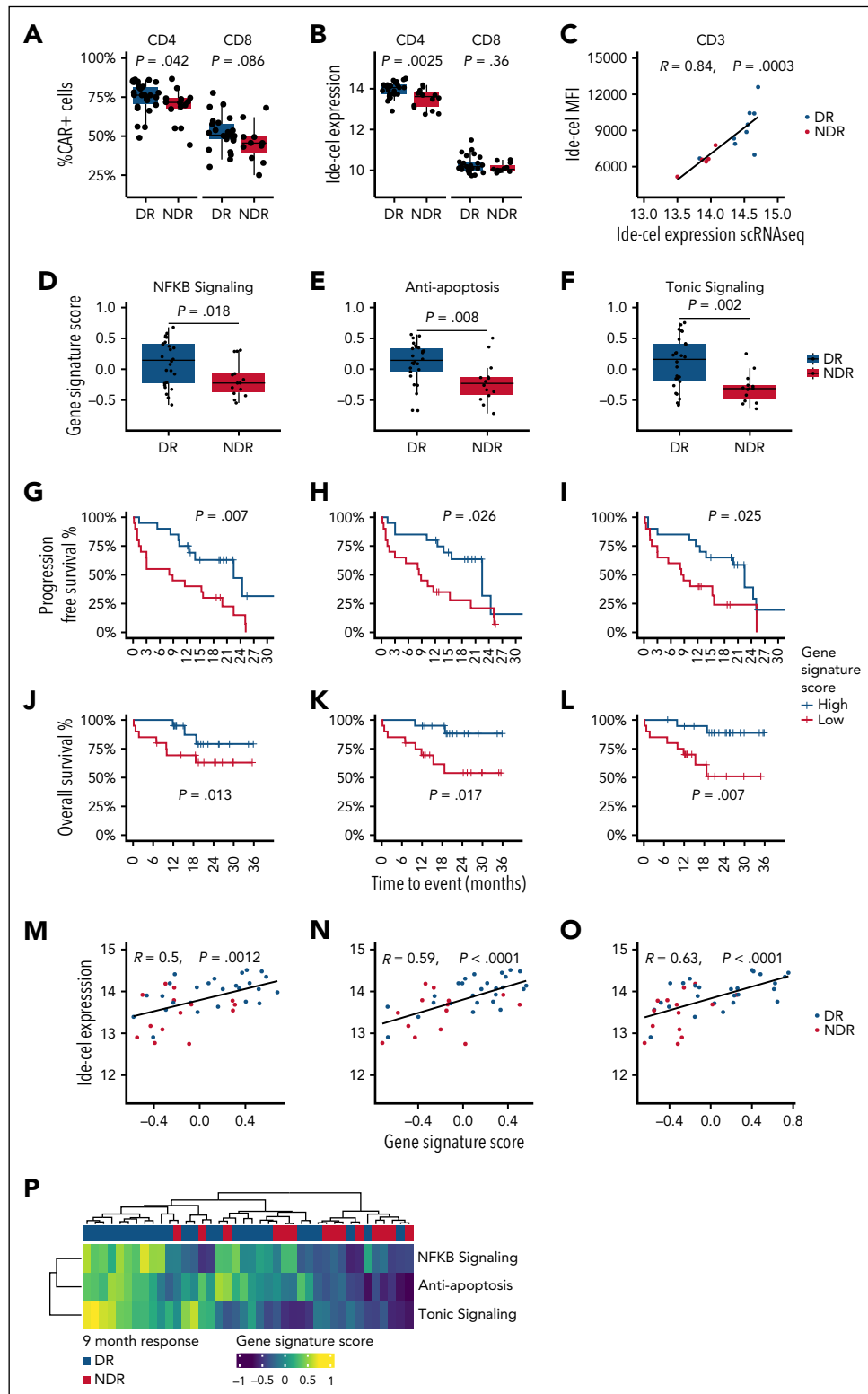


Figure 2. Ide-cel expression reflects NF-κB and prosurvival programs that influence efficacy. (A) Percentage of CAR⁺ cells in ide-cel IP between responders in CD4 cells of 40 patients and CD8 cells of 36 patients (those with <50 cells removed). (B) Pseudobulk expression of the ide-cel construct in CAR⁺ ide-cel IP cells between responders in CD4 cells of 40 patients and CD8 cells of 36 patients (those with <50 cells removed). *P* values from unpaired Wilcoxon rank-sum test are shown in panels A-B. (C) Spearman correlation between ide-cel surface protein expression by flow cytometry and pseudobulk transcriptional expression from scRNA-seq of all IP cells (*n* = 16 patients with matched scRNA-seq and flow cytometry data). (D-F) Expression of NF-κB signaling, antiapoptosis, and tonic signaling signatures via pseudobulk aggregation of gene expression followed by gsva in CAR⁺ CD4 cells of 40 patients between DR and NDR. *P* values from unpaired Wilcoxon rank-sum test are shown. (G-L) Association of CAR⁺ CD4 NF-κB signaling, antiapoptosis, and tonic signaling signatures as calculated in panels D-F above or below (high vs low) the median expression of the cohort and PFS and OS (*n* = 40 patients). *P* values from log-rank test are shown. (M-O) Spearman correlation between NF-κB signaling, antiapoptosis, and tonic signaling signatures as calculated in panels D-F, and pseudobulk expression of the ide-cel construct in CAR⁺ CD4 cells (*n* = 40 patients) with points labeled by 9-month response. (P) Heat map depicting the expression of NF-κB signaling, antiapoptosis, and tonic signaling signatures as calculated in panels D-F, and pseudobulk expression of ide-cel CAR construct in all CAR⁺ CD4 cells (*n* = 40 patients) and their association with durable response.

Table 1. Patient baseline characteristics stratified by ide-cel IP NF-κB signature

Category	N = 40	High NF-κB signature (N = 20)	Low NF-κB signature (N = 20)	P value
Age, y, median (range)	67 (43-81)	68 (43-81)	66 (51-80)	.2
Race, n (%)				.3
White	31 (78)	16 (80)	15 (75)	
Black	7 (18)	4 (20)	3 (15)	
Other	2 (4)	0 (0)	2 (10)	
Sex, n (%)				.2
Male	22 (55)	9 (45)	13 (65)	
Female	18 (45)	11 (55)	7 (35)	
ECOG before LD chemo, n (%)				.2
0	23 (57)	10 (50)	13 (65)	
1	14 (35)	7 (35)	7 (35)	
2	3 (8)	3 (15)	0 (0)	
Time from diagnosis to CAR-T, y, median (range)	6.4 (2-12)	6.7 (2-12)	6.2 (2-12)	.97
Prior lines of therapy, n, median (range)	6 (4-13)	6 (4-9)	6 (4-13)	.4
Prior autologous stem cell transplantation, n (%)	25 (63)	9 (45)	16 (80)	.02
Time from autologous transplant to CAR-T, y, median (range)	5.3 (0.9-10.6)	6.6 (2.4-8.2)	4.1 (0.9-10.6)	.2
Penta-refractory status, n (%)	15 (38)	5 (25)	10 (50)	.1
R-ISS stage before LD chemotherapy, n (%)				.4
I	4 (10)	3 (15)	1 (5)	
II	19 (48)	10 (50)	9 (45)	
III	2 (5)	0 (0)	2 (10)	
Missing	15 (37)	7 (35)	8 (40)	
High-risk FISH, n (%)	8 (20)	4 (20)	4 (20)	1.0
del(17p) or t(4:14), n (%)	1 (3)	1 (5)	0 (0)	.3
t(14:16), n (%)	3 (8)	1 (5)	2 (10)	.5
Gain/amplification of Ch1q, n (%)	15 (37)	7 (35)	8 (40)	.7
Bridging therapy, n (%)	23 (58)	13 (65)	10 (50)	.4
Pretreatment extramedullary disease, n (%)	6 (15)	4 (20)	2 (10)	.4
Prior BCMA-directed therapy, n (%)	8 (20)	4 (20)	4 (20)	1.0
Ide-cel dose, ×10 ⁶ CAR ⁺ cells, median (range)	417.6 (316.5-456.7)	426.1 (321.1-456.7)	413.1 (316.5-454)	.5
Absolute lymphocyte count at apheresis, ×10 ³ /μL, median (range)	0.67 (0.13-3.19)	0.84 (0.24-3.19)	0.35 (0.13-2.71)	.2
CAR-HEMATOTOX score at apheresis, n (%)				.5
HT low risk (score 0-1)	18 (45)	8 (40)	10 (50)	
HT high risk (score ≥2)	22 (55)	12 (60)	10 (50)	
CD4:CD8 ratio at apheresis, median (range)	0.725 (0.11-5.43)	0.67 (0.24-5.43)	1.06 (0.11-2.47)	.8
Lymphocyte-to-neutrophil ratio at apheresis, cells per μL, median (range)	0.24 (0.07-4.5)	0.3 (0.08-4.5)	0.2 (0.07-1.4)	.2

Bold indicates the only *P* value to reach the significance threshold of <0.05. CRP, C-reactive protein; ECOG, Eastern Cooperative Oncology Group; FISH, fluorescence in situ hybridization; HT, high-throughput; LD, lymphodepleting; R-ISS, Revised International Staging System.

Table 1 (continued)

Category	N = 40	High NF-κB signature (N = 20)	Low NF-κB signature (N = 20)	P value
sBCMA at apheresis, ng/mL, median (range)	93.4 (4.2-1353.8)	110.5 (8.5-1321)	85.2 (4.2-1353.8)	1.0
Pre-CAR-T burden in marrow (≥ 50% plasma cells), n (%)				.05
Yes	12 (30)	8	19	
No	25 (63)	3 (17)	10 (53)	
Poor sample/not performed	3 (7)	15 (83)	9 (47)	
β ₂ -microglobulin before LD chemotherapy, mg/L, median (range)	3.3 (1.5-9.3)	3.1 (1.9-6.7)	3.4 (1.5-9.3)	.2
Ferritin level before LD chemotherapy, μg/L, median (range)	219 (20-4960)	151 (27-3245)	235 (20-4960)	.4
CRP level before LD chemotherapy, mg/dL, median (range)	0.4 (0.04-13.9)	0.32 (0.04-8.87)	0.37 (0.06-10.6)	.4
sBCMA at LD, ng/mL, median (range)	119.4 (4.4-2708.3)	190.8 (9.9-2708.3)	110.3 (4.5-1150.4)	.5
Duration of follow-up of living patients, mo, median (range)	28.9 (7.1-44.1)	27.2 (12.1-44.1)	29.2 (7.1-40.4)	.8

Bold indicates the only P value to reach the significance threshold of <0.05. CRP, C-reactive protein; ECOG, Eastern Cooperative Oncology Group; FISH, fluorescence in situ hybridization; HT, high-throughput; LD, lymphodepleting; R-ISS, Revised International Staging System.

greater proportion of patients in the low NF-κB signature group who had undergone prior ASCT (80% vs 45%; $P = .02$), with these transplants also being performed more recently relative to apheresis (median interval, 4.1 years in low NF-κB vs 6.6 years in high NF-κB); however, this did not reach statistical significance.

Given the association of longitudinal sBCMA with disease burden, we jointly modeled it and the NF-κB CD4 convergence score in Cox analyses. The convergence score showed a consistent protective association and remained independently associated with OS at early postinfusion landmarks after adjusting for concurrent sBCMA, whereas both variables contributed to PFS at selected time points (notably day 28), with sBCMA more strongly associated with PFS at later landmarks (supplemental Figures 21-24).

Detailed flow cytometric phenotypic analysis of the apheresis product sent for manufacture, stratified according to the NF-κB signature observed in the paired IP samples (supplemental Figures 25 and 26), revealed differences at baseline that might contribute to the generation of a product having higher NF-κB activity. Specifically, patients with a high NF-κB signature in the IP had a more favorable baseline T-cell phenotype, including increased proportion of CD4 central memory T cells (CD45RO⁺ CCR7⁺) (Figure 3A), reduced expression of CD39 and PD-1 (Figure 3B-C), and higher levels of TCF1 (Figure 3D), indicative of CD4 T-cell stemness and persistence. Further, patients with high NF-κB signature in the IP also were more likely to have baseline CD4 cells that did not express checkpoint receptors (PD-1[−] TIGIT[−] LAG-3[−]; Figure 3E), whereas patients with low NF-κB were more likely to express at least 1 of these ligands (Figure 3F). There were also linear relationships between these markers and the NF-κB signature in the IP

(Figures 3G-L). Similarly, in the CD8 population, patients with low NF-κB were more likely to possess T cells that expressed at least 1 checkpoint receptor (supplemental Figure 26). No significant differences were observed in the proportions of regulatory T cells, myeloid-derived suppressor cells, or natural killer cells between patients with high NF-κB and those with low NF-κB that could explain the differences in these signatures (supplemental Figures 27 and 28).

As CAR-T cells are manufactured from peripheral blood T cells, we hypothesized that bone marrow TME may inform circulating T cells and, subsequently, the CAR-T product. Analyses of paired samples from patients with R/R MM confirmed a strong correlation between T-cell phenotypes by flow cytometry in peripheral blood mononuclear cell and bone marrow samples (supplemental Figures 29 and 30). Furthermore, T-cell flow cytometry phenotyping and gene expression profiling using the NanoString PanCancer Immune Profiling platform of a broader set of bone marrow samples from patients with R/R MM (total N = 30, with 9 paired samples), revealed that NF-κB pathway activity in the TME correlated with CD4 T-cell phenotypes observed in the IP, where there was a negative correlation between NF-κB signaling and PD-1 expression and the expression of checkpoint receptors (Figure 3M-O). Collectively, these observations suggest that baseline T-cell fitness in the TME and in circulation shapes NF-κB signaling in the CAR-T cell product, supporting NF-κB as a potential driver of CAR-T cell function.

Transient NF-κB inhibition impairs CAR-T cytotoxicity and cytokine production

To assess the contribution of NF-κB signaling to CAR-T cell effector function, anti-BCMA CAR-T cells were transiently treated with BMS-345541 for 2 hours, followed by washout

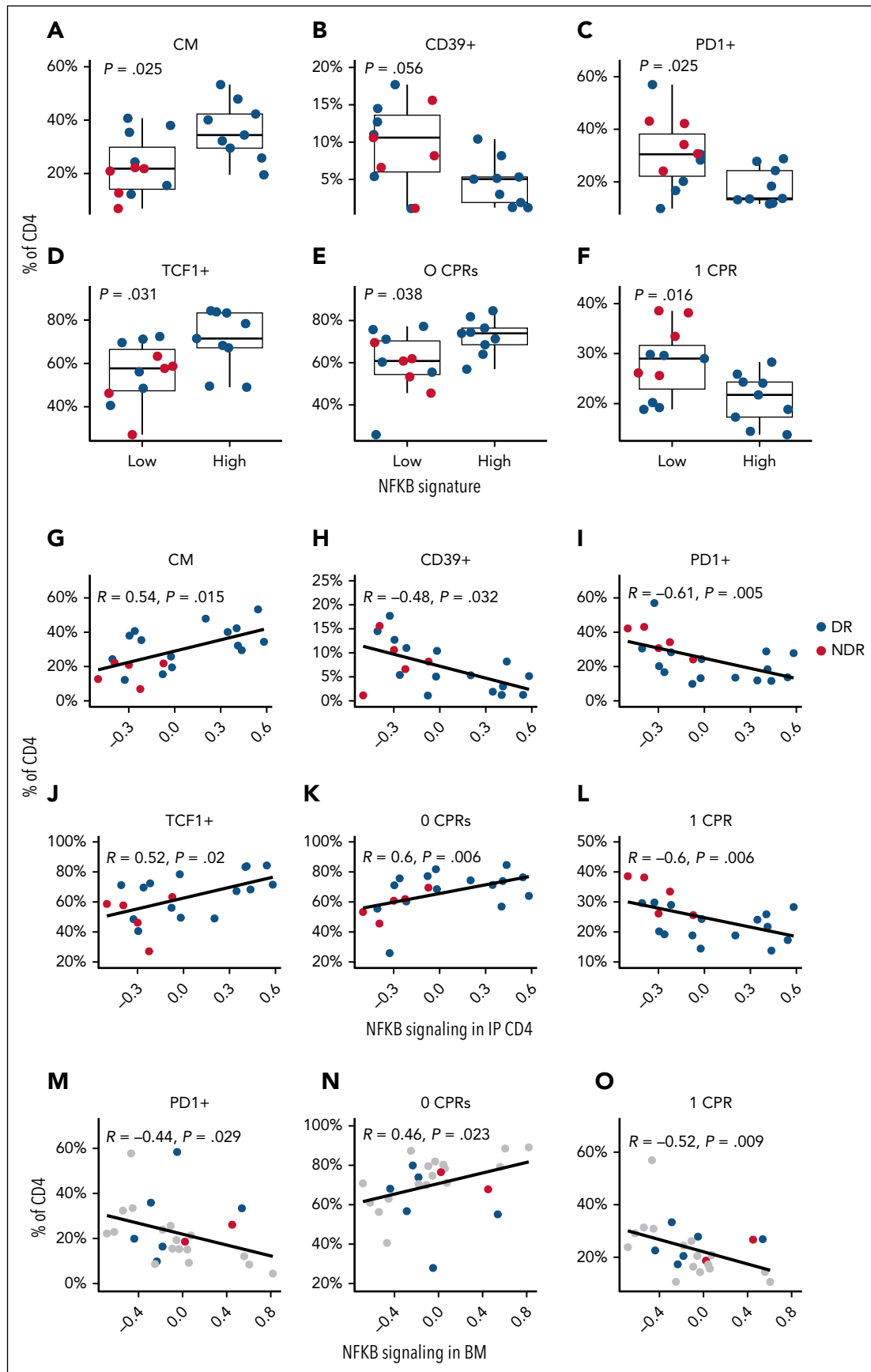


Figure 3.

before assay initiation (Figure 4A-B; also see “Methods”). The inhibition of NF- κ B activation was confirmed by reduced phosphorylation of the NF- κ B p65 subunit compared with control conditions (supplemental Figures 31 and 32).

CAR-T cytotoxicity was first evaluated after a single round of antigen stimulation at increasing effector-to-target (E:T) ratios. Transient NF- κ B inhibition was associated with reduced CAR-T-mediated cytotoxicity compared with control CAR-T cells (Figure 4A-B; also see “Methods”). Cytokine secretion was assessed after 24 hours of coculture with BCMA-expressing NCI-H929 target cells. NF- κ B-inhibited CAR-T cells showed reduced production of IL-2, tumor necrosis factor α , and IL-6 across E:T ratios (Figure 4C). These data support a role for canonical NF- κ B signaling in CAR-T cytotoxicity and cytokine production after antigen stimulation.

We next assessed whether this impairment persisted under repeated antigen exposure (Figure 4D-E). CAR-T cells were subjected to serial rounds of antigen stimulation, with transient NF- κ B inhibition applied before each stimulation cycle, and cytotoxic activity was assessed across E:T ratios at defined time points. Across repeated stimulation cycles, transiently inhibited CAR-T cells exhibited reduced cytotoxic activity compared with control CAR-T cells (Figure 4E), indicating impaired effector function during repeated antigen exposure.

Transient NF- κ B inhibition impairs CAR-T-mediated tumor control over time, independent of changes in CAR-T proliferation, viability, or phenotype

To further assess CAR-T cell function under conditions of repeated antigen stimulation, real-time killing assays were performed using live-cell imaging (as outlined in Figure 5A). CAR-T cells were cocultured with GFP-expressing NCI-H929 target cells, and tumor burden was monitored longitudinally using the IncuCyte system. Transient NF- κ B inhibition of CAR-T cells exhibited reduced tumor control over time compared with control CAR-T cells (Figure 5A-B), and consistent results were observed in additional independent healthy donors (supplemental Figure 33A-B), without NF- κ B inhibition affecting tumor-only controls (supplemental Figure 33C-D).

To determine whether impaired tumor control after transient NF- κ B inhibition was associated with changes in CAR-T cell proliferative capacity or phenotype, CAR-T cells were analyzed across repeated rounds of antigen stimulation (Figure 5C). Cell division, assessed by CellTrace Violet dilution, revealed comparable proliferation between transiently inhibited CAR-T cells and control CAR-T cells (Figure 5D; supplemental Figure 34).

CAR-T cell viability was also similar between groups (Figure 5E).

The frequencies of CD39⁺, PD-1⁺, and TCF1⁺ CAR-T cells were comparable between conditions across stimulation cycles, as were the distributions of effector memory and central memory subsets (Figure 5F; supplemental Figure 35). The overall proportion of CAR-expressing T cells remained unchanged after transient NF- κ B inhibition (supplemental Figure 36). Together, these data indicate that transient NF- κ B inhibition primarily affects CAR-T cell effector function rather than proliferation, viability, or phenotypic composition.

Collectively, these data support a role for NF- κ B signaling in CAR-T cell effector function, influencing both cytotoxic activity and cytokine secretion, and support NF- κ B activity as a functional determinant of CAR-T cell responses.

Discussion

This scRNA-seq analysis of standard-of-care ide-cel IP identified biological features of ide-cel associated with durable clinical responses in multiple myeloma. The patient cohort mirrored the demographics of other real-world cohorts, reinforcing the clinical relevance of the findings. CAR density and 4-1BB tonic signaling coincided with a canonical NF- κ B-associated prosurvival program that was most evident in CAR⁺ CD4 cells and tracked with longer duration of response. Concordant signatures (NF- κ B signaling, antiapoptosis, and tonic signaling) and their alignment with higher CAR transgene expression suggest a product state that is characterized by preserved CD4 helper function and reduced susceptibility to activation-induced cell death, which may support persistence and coordination of CD8 effector activity. Notably, NF- κ B signaling observed in CAR-T IP from patients with durable responses is concordant with a c-REL-dependent transcriptional program,²¹ and inhibition of the c-REL-activating kinase I κ B kinase β impairs CAR-T-directed cytotoxicity and cytokine production. Thus, NF- κ B/c-REL-linked programs may mark a cellular state associated with CAR-T product fitness and clinical benefit.

In ide-cel products, there was relative uniformity in broad cellular composition, with minimal associations of single-cell clusters and efficacy, not unlike the findings with CD19 CAR-T.²³ Uniformity is not unexpected for a commercial product approved by the US Food and Drug Administration where release criteria focus on sterility, viability, transduction efficiency, and limited functional assays rather than detailed phenotypic or transcriptional profiling. However, products that

Figure 3. NF- κ B signaling in CAR⁺ CD4 cells of ide-cel IP is determined by CD4 T-cell differentiation and exhaustion states of apheresis starting material. Percentage of apheresis-derived CD4⁺ T cells with the following phenotypes: (A) central memory (CD45RO⁺CCR7⁺), (B) CD39 expression, (C) PD-1 expression, (D) TCF1 expression, (E) absence of checkpoint receptors (CPRs) (PD-1[−]TIGIT[−]LAG-3[−]), and (F) expression of a single CPR. Data are stratified by the ide-cel IP NF- κ B signaling signature, quantified in CAR⁺ CD4⁺ cells using pseudobulk scRNA-seq gene expression followed by gsva in 20 patients with available matched scRNA-seq and flow cytometry data. Patients were dichotomized into high vs low signature expression based on the cohort median. P values from unpaired Wilcoxon rank-sum test are reported. (G-L) Spearman correlation between the percentage of collected apheresis CD4 cells that have a (G) central memory (CD45RO⁺ CCR7⁺) phenotype, (H) express CD39, (I) express PD-1, (J) express TCF1, (K) express zero CPRs (PD-1[−] TIGIT[−] LAG-3[−]), and (L) express 1 CPR in 20 patients with available matched scRNA-seq and flow cytometry data vs ide-cel IP NF- κ B signaling signature as calculated in panels A-F. (M-O) Spearman correlation between the percentage of collected CD138-depleted BM aspirate TME CD4 cells that (M) express PD-1, (N) express zero CPRs (PD-1[−] TIGIT[−] LAG-3[−]), and (O) express 1 CPR vs NF- κ B signaling measured via genes captured in the NanoString PanCancer Immune Profiling gene expression panel in 30 patients (9 with matched scRNA-seq data). Blue points indicate a durable response of at least 9 months to ide-cel, red points indicate the patient had a nondurable response, and gray points are additional relapsed/refractory myeloma patients with evaluable TME that were not part of the ide-cel IP cohort. BM, bone marrow.

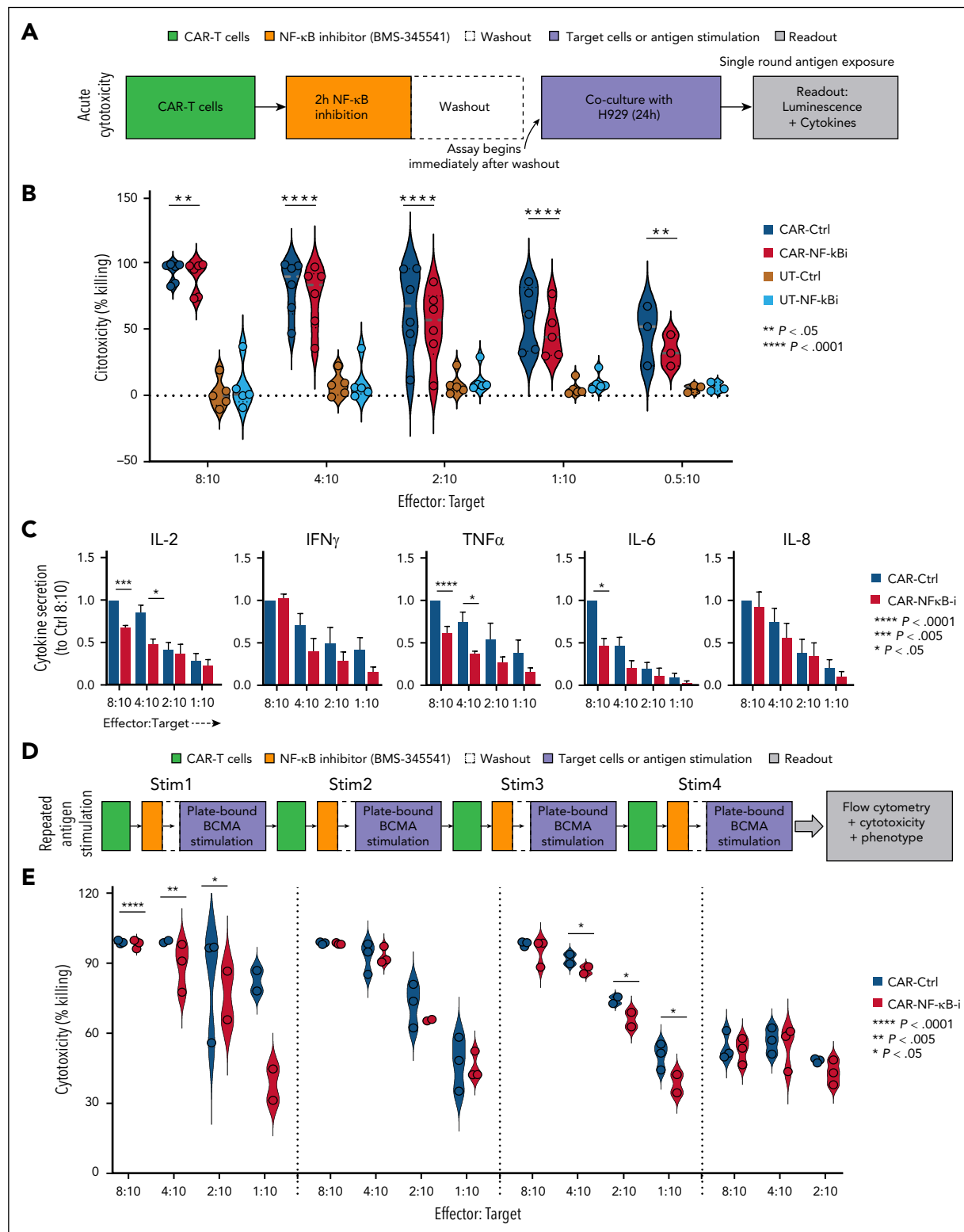


Figure 4. Transient NF- κ B inhibition compromises CAR-T cytotoxic function across E:T ratios and repeated antigen stimulation. (A) Schematic representation of the experimental workflow illustrating transient NF- κ B inhibition using BMS-345541. (B) Cytotoxic activity of control CAR-T cells, NF- κ B-inhibited CAR-T cells, and untransduced T cells measured across indicated E:T ratios after a single stimulation. (C) Cytokine secretion measured after 24-hour coculture with target cells at the indicated E:T ratios after a single stimulation. (D) Schematic of the repeated antigen stimulation assay in which CAR-T cells underwent serial plate-bound BCMA stimulations, followed by functional and phenotypic analyses. (E) Cytotoxic activity of control and NF- κ B-inhibited CAR-T cells measured across E:T ratios after sequential rounds of antigen stimulation from Stim1 to Stim4. Data are presented as mean $\pm$ SEM from independent donors. Panel B includes 6 donors, panel C includes 3 donors, and panel E includes 3 donors. Statistical significance was determined using 2-way mixed-effects ANOVA, as indicated in the figure. ANOVA, analysis of variance; Ctrl, control; SEM, standard error of the mean; Stim, stimulation.

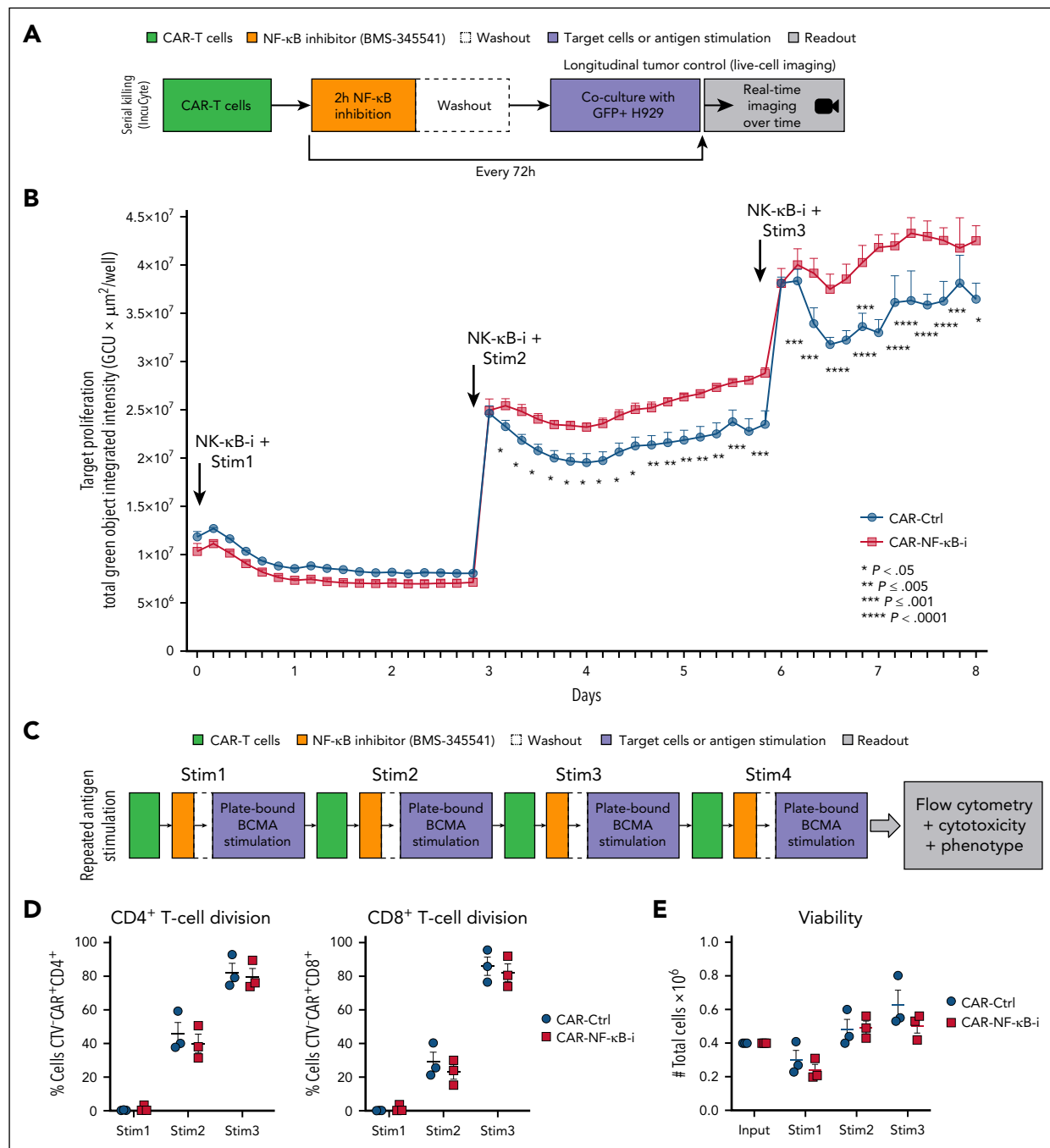


Figure 5. Transient NF-κB inhibition impairs longitudinal CAR-T tumor control despite preserved proliferation, viability, and differentiation state. (A) Experimental schematic illustrating transient NF-κB inhibition and longitudinal assessment of tumor control by IncuCyte live-cell imaging. Tumor cell killing was continuously monitored over time, with challenges occurring at 72-hour intervals. (B) Longitudinal quantification of target cell proliferation measured by total integrated GFP intensity during repeated tumor cell challenges. Arrows indicate the timing of each antigen challenge and NF-κB inhibition. (C) Schematic of the repeated plate-bound BCMA antigen stimulation protocol used for downstream analyses. (D) Quantification of CD4⁺ and CD8⁺ CAR-T cell division and (E) CAR-T cell viability after each round of plate-bound BCMA stimulation. (F) Phenotypic characterization of CAR-T cells after repeated stimulation, including expression of CD39, TCF1, and PD-1, as well as the distribution of effector memory and central memory T-cell subsets. Data are presented as mean ± SEM from 3 independent donors. Statistical significance was determined using 2-way ANOVA, as indicated in the figure. Experiments shown in panels D-F were performed using parallel cultures of plate-bound antigen-stimulated CAR-T cell and do not correspond to the same cells analyzed in the live-cell imaging assays. ANOVA, analysis of variance; Ctrl, control; SEM, standard error of the mean; Stim, stimulation.

meet release specifications can still harbor biologically meaningful heterogeneity, as demonstrated in the analyses of commercial ide-cel products showing variability in manufacturing attributes and clinical outcomes.²⁴

We found that CAR-T NF-κB pathway activity aligned with baseline T-cell features previously linked to CAR-T fitness and efficacy, including more stem and central memory cells and lower expression of checkpoint receptors,^{25,26} but not the

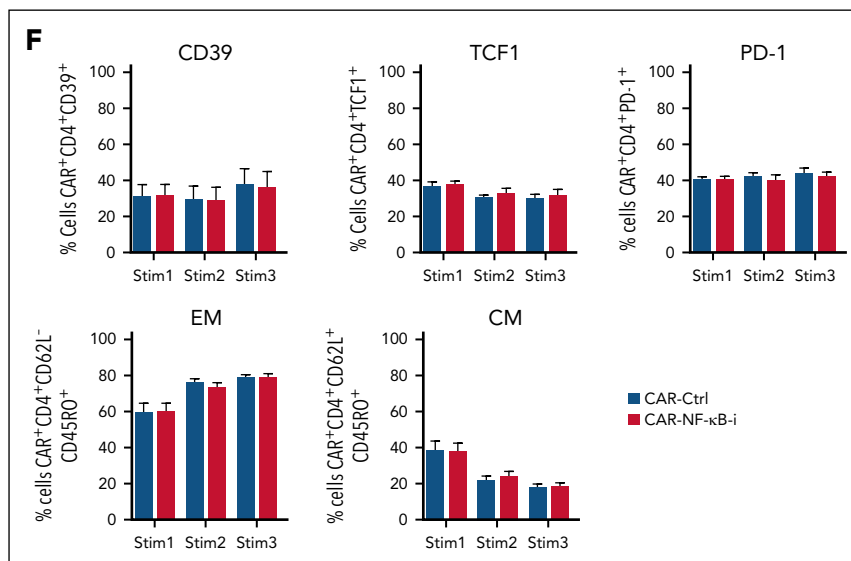


Figure 5 (continued)

CD4:CD8 ratio. The link between CAR-T NF-κB signaling and premanufacture immune phenotypes in matched apheresis and TME samples (Figures 3-4), together with the coupling of CAR density, tonic signaling, and prosurvival transcriptional programs in CAR⁺ CD4 cells (Figure 2), support that NF-κB signaling represents an integrated product fitness state.

Our functional experiments further support a role for NF-κB signaling in CAR-T cell effector activity. Transient pharmacologic inhibition of canonical NF-κB signaling consistently reduced CAR-T-mediated cytotoxicity and cytokine secretion against BCMA-expressing tumor cells. However, NF-κB inhibition did not substantially alter CAR-T cell proliferation, viability, or phenotypic composition across repeated rounds of antigen stimulation. Taken together, our data indicate that NF-κB signaling may not primarily drive rapid phenotypic remodeling but instead reflect a broader transcriptional state associated with T-cell functional competence. In this context, NF-κB activity within the ide-cel product may represent both a signaling axis that contributes to CAR-T cytotoxic function and a broader marker of global T-cell fitness that ultimately influences therapeutic durability. The concordance between IP signatures and matched apheresis T-cell differentiation and exhaustion phenotypes indicates that premanufacture T-cell fitness seeds product-intrinsic quality. Importantly, the link between NF-κB signaling and T-cell fitness was also detected in the myeloma TME, suggesting that tumor-intrinsic features may drive poor T-cell fitness to compromise CAR-T efficacy. This creates opportunities upstream of manufacturing, optimizing apheresis timing, selecting therapies that spare T-cell fitness during bridging, and considering short precollection washout windows where clinically feasible. Such strategies could shift the starting composition toward NF-κB-high, central memory-like, and less-exhausted phenotypes associated with durability.

Our findings also support developing manufacturing strategies that preserve central memory-like phenotypes and avoid excessive ex vivo stimulation and expansion that are associated with worse efficacy with CAR-T for lymphoma.²³ Practical levers

include gentler activation conditions, IL-7/IL-15 support, and tuning CAR expression to mitigate excessive signaling.

Although tonic signaling has been linked to premature exhaustion in CAR-T cells bearing CD28 costimulatory domains, this concept may not be generalized across CAR architectures. Prior studies in 4-1BB-based systems have yielded mixed results, including reports that excessive 4-1BB tonic signaling in experimental donor-derived models may impair persistence, whereas others suggest that 4-1BB-associated tonic signaling can promote CAR-T efficacy.^{16,27-30} In our study, analysis of clinical patient-derived ide-cel IP supports an association between an NF-κB-associated program and durable response.

Mechanisms of resistance to anti-BCMA CAR-T therapy are multifactorial, and we previously demonstrated that discordantly low sBCMA in the setting of high disease burden identifies bona fide antigen loss associated with inferior outcomes.¹⁸ Here, we found that sBCMA and the NF-κB CD4 convergence score were independently associated with PFS and OS across multiple time points, supporting the interpretation that this product-intrinsic fitness state captures a complementary and nonredundant axis of therapeutic durability. Limitations to this study include its retrospective nature and the lack of an external validation cohort. To our knowledge, this study is the first to report scRNA-seq analysis of a commercial myeloma CAR-T product. Although baseline clinical characteristics were otherwise comparable, a higher proportion of patients in the NF-κB_{low} group had prior ASCT (with transplants occurring a median of several years before apheresis), underscoring that treatment history may shape premanufacture T-cell fitness and that these correlative associations should be interpreted as hypothesis-generating. We acknowledge that the assessment of tumor BCMA expression at relapse remains a critical unmet need and an important limitation here and shared broadly across real-world studies. We also recognize that these findings are restricted to ide-cel and thus may not be generalizable to other BCMA-directed products, costimulatory domains, or

alternative targets. Pathway-level signatures were evaluated as complementary biological context across several curated and overlapping gene sets; these analyses should be interpreted as supportive rather than definitive on their own. However, convergent functional assays confirmed the importance of NF- κ B signaling on ide-cel CAR-T function.

Collectively, our results support that NF- κ B in the ide-cel product marks a signaling axis affecting CAR-T function, primarily cytotoxicity, and more broadly, NF- κ B activity represents a global marker of T-cell fitness present before CAR-T manufacture. The concordance between IP signatures and paired apheresis/TME phenotypes underscores an upstream opportunity to identify T cells with superior fitness at apheresis to generate more durable CAR-T products. These insights also support strategies to preserve central memory-like states, avoid excessive ex vivo stimulation, and consider precollection interventions to improve T-cell quality. Mechanistically, our data suggest that CAR density and 4-1BB tonic signaling are associated with an NF- κ B-linked prosurvival program in CD4 CAR-T cells that is associated with prolonged clinical benefit. Validation using the c-REL-dependent program further supports NF- κ B/c-REL as a central node of CAR-T persistence and function. Together, our data provide a framework for optimizing patient selection, bridging therapy, and manufacturing approaches to sustain durable responses to CAR-T in myeloma.

Acknowledgments

The authors thank the Flow Cytometry Core, the Molecular Genomics Core, and the Tissue Core of Moffitt Cancer Center for providing the services.

This work was supported, in part, by the American Cancer Society, internal funding, the National Cancer Institute, National Institutes of Health (P30CA076292 [J.L.C., principal investigator] and R01CA244328 [F.L.L., multiple principal investigator]), and generous donations from the Hyer family, the Hano family, and the Thiel family. F.L.L. is supported, in part, by the Leukemia and Lymphoma Society as a Clinical Scholar (2334-21). J.L.C. is supported by the Cortner-Couch Endowed Chair for Cancer Research at the University of South Florida Morsani College of Medicine.

Authorship

Contribution: J.D.N. analyzed and curated biostatistical data and wrote the first and final drafts of the manuscript; B.C.P. performed biological experiments and analyzed data; X.S. contributed to biostatistical input and review; M.A.M. contributed to immunophenotyping and data generation; J.L.C. critically reviewed and edited the manuscript; F.L.L. and C.L.F. contributed to experimental design, oversight, data analysis, curation, and writing of all manuscript drafts, and jointly supervised the work; and all authors reviewed the data, contributed to the manuscript writing, and approved the submitted version of the manuscript.

Conflict-of-interest disclosure: O.A.C.P. reports honoraria for consulting from Legend Biotech USA Inc and Bristol Myers Squibb (BMS). H.L. reports a role with the speakers' bureau for Sanofi. M.A. reports consulting fees from Janssen, BMS, and Sanofi; and research support from BMS. T.N. reports clinical trial support to the institution from Novartis; clinical trial support, consisting of drug-only supply to the institution, from Karyopharm; consultancy fees from ImmunoGen; and advisory board participation for Medexus. K.H.S. reports consultancy, advisory, and/or speaker roles with Adaptive Biotech, Janssen, BMS, Takeda,

Sanofi, and GlaxoSmithKline; research funding from Karyopharm and AbbVie; and clinical trial funding from BMS, Amgen, and Janssen. R.C.B. reports honoraria for consulting from BMS, Janssen, and Pfizer; and research funding from BMS, Janssen, AbbVie, Regeneron, and Karyopharm. B.J.B. reports speakers' bureau participation for Sanofi Pharmaceuticals. A.F.G.-C. serves on advisory boards for Janssen and Sanofi; and reports a role with the speakers' bureau for Sanofi. D.K.H. reports research funding from BMS, Karyopharm, and Adaptive Biotech; consulting or advisory roles for BMS, Janssen, Pfizer, and Karyopharm; and support from the Pentecost Family Myeloma Research Center. F.L.L. reports scientific advisory roles with and consulting fees from A2, Allogene, Amgen, Bluebird Bio, BMS, Calibr, Caribou, Cowen, EcoR1, Gerson Lehrman Group, Iovance, Kite Pharma, Janssen, Legend Biotech, Novartis, Sana, Umoja, and Pfizer; membership of the data and safety monitoring board for the National Cancer Institute Safety Oversight CAR T-cell Therapies Committee; research contracts and grants from Kite Pharma, Allogene, CERo Therapeutics, Novartis, Bluebird Bio, 2SeventyBio, and BMS; patents, royalties, and other intellectual property related to several patents held by the institution in his name (unlicensed) in the field of cellular immunotherapy; and educational or editorial activities for Aptitude Health, American Society of Hematology, BioPharm Communications CARE Education, Clinical Care Options Oncology, Imedex, and the Society for Immunotherapy of Cancer. C.L.F. reports honoraria for consulting from BMS/Celgene, ONK Therapeutics, and Janssen; and research funding from BMS and Janssen. The remaining authors declare no competing financial interests.

ORCID profiles: B.C.P., [0000-0002-8228-0194](https://orcid.org/0000-0002-8228-0194); L.A.C.D., [0009-0007-8037-9784](https://orcid.org/0009-0007-8037-9784); J.C.L., [0009-0001-3928-4018](https://orcid.org/0009-0001-3928-4018); H.L., [0000-0002-9945-4381](https://orcid.org/0000-0002-9945-4381); T.N., [0000-0002-2621-7924](https://orcid.org/0000-0002-2621-7924); K.H.S., [0000-0003-4178-8888](https://orcid.org/0000-0003-4178-8888); R.C.B., [0000-0003-4538-4733](https://orcid.org/0000-0003-4538-4733); A.F.G.-C., [0000-0003-0999-1771](https://orcid.org/0000-0003-0999-1771); D.K.H., [0000-0002-9813-3578](https://orcid.org/0000-0002-9813-3578); F.P., [0000-0002-3744-8672](https://orcid.org/0000-0002-3744-8672); C.C.L., [0000-0002-4506-6244](https://orcid.org/0000-0002-4506-6244); J.M.B., [0000-0001-5871-686X](https://orcid.org/0000-0001-5871-686X); F.L.L., [0000-0001-9063-6691](https://orcid.org/0000-0001-9063-6691); C.L.F., [0000-0002-0662-6480](https://orcid.org/0000-0002-0662-6480).

Correspondence: Ciara L. Freeman, Department of Blood and Marrow Transplant and Cellular Immunotherapy, H. Lee Moffitt Cancer Center & Research Institute, 12902 USF Magnolia Dr, Tampa, FL 33612; email: ciara.freeman@moffitt.org; and Frederick L. Locke, Department of Blood and Marrow Transplant and Cellular Immunotherapy, H. Lee Moffitt Cancer Center & Research Institute, 12902 USF Magnolia Dr, Tampa, FL 33612; email: frederick.locke@moffitt.org.

Footnotes

Submitted 29 September 2025; accepted 20 May 2026; prepublished online on *Blood* First Edition 11 June 2026. <https://doi.org/10.1182/blood.2025031843>.

*J.D.N. and B.C.P. contributed equally to this study.

†F.L.L. and C.L.F. jointly supervised this study and are joint corresponding authors.

Presented in oral abstract form at the 67th annual meeting of the American Society of Hematology, Orlando, FL, 7 December 2025.

Single-cell RNA sequencing data have been deposited in the National Center for Biotechnology Information Gene Expression Omnibus database (accession number GSE277937).

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

The publication costs of this article were defrayed in part by page charge payment. Therefore, and solely to indicate this fact, this article is hereby marked "advertisement" in accordance with 18 USC section 1734.

REFERENCES

- Munshi NC, Anderson LD, Shah N, et al. Idecabtagene vicleucel in relapsed and refractory multiple myeloma. *N Engl J Med*. 2021;384(8):705-716.
- Hansen DK, Peres LC, Dima D, et al. Comparison of standard-of-care ideoabtagene vicleucel and ciltacabtagene autoleucel in relapsed/refractory multiple myeloma. *J Clin Oncol*. 2025;43(13):1597-1609.
- Rodriguez-Otero P, Ailawadhi S, Arnulf B, et al. Ide-cel or standard regimens in relapsed and refractory multiple myeloma. *New Engl J Med*. 2023;388(11):1002-1014.
- Berdeja JG, Madduri D, Usmani SZ, et al. Ciltacabtagene autoleucel, a B-cell maturation antigen-directed chimeric antigen receptor T-cell therapy in patients with relapsed or refractory multiple myeloma (CARTITUDE-1): a phase 1b/2 open-label study. *Lancet*. 2021;398(10297):314-324.
- Lin Y, Martin TG, Usmani SZ, et al. CARTITUDE-1 final results: phase 1b/2 study of ciltacabtagene autoleucel in heavily pretreated patients with relapsed/refractory multiple myeloma. *J Clin Oncol*. 2023;41(16_suppl):8009.
- San-Miguel J, Dhakal B, Yong K, et al. Ciltacel or standard care in lenalidomide-refractory multiple myeloma. *N Engl J Med*. 2023;389(4):335-347.
- Jagannath S, Martin TG, Lin Y, et al. Long-term (≥ 5 -Year) remission and survival after treatment with ciltacabtagene autoleucel in CARTITUDE-1 patients with relapsed/refractory multiple myeloma. *J Clin Oncol*. 2025;43(25):2766-2771.
- Alsina M, Shah N, Raje NS, et al. Updated results from the phase I CRB-402 study of anti-Bcma CAR-T cell therapy bb21217 in patients with relapsed and refractory multiple myeloma: correlation of expansion and duration of response with T cell phenotypes. *Blood*. 2020;136(suppl 1):25-26.
- Daniela GMT, Amanda JO, Bianca von S, et al. Understanding T cell phenotype for the design of effective chimeric antigen receptor T cell therapies. *J ImmunoTherapy Cancer*. 2021;9(5):e002555.
- Freeman CL, Abraham-Miranda J, Menges M, et al. Enhanced CAR-T function and mitochondrial fitness from earlier unfractionated stem cell product in multiple myeloma. *Mol Ther*. 2025;33(8):3576-3589.
- Locke FL, Rossi JM, Neelapu SS, et al. Tumor burden, inflammation, and product attributes determine outcomes of axicabtagene ciloleucel in large B-cell lymphoma. *Blood Adv*. 2020;4(19):4898-4911.
- Julie Rytlewski JF, Mertz D, Freeman C, Manier S, Shah N, Campbell T. Correlative analysis to define patient profiles associated with manufacturing and clinical endpoints in relapsed refractory multiple myeloma patients treated with idecabtagene vicleucel (ide-cel; bb2121), an anti-BCMA CAR T-cell therapy. 2022;6(suppl):759-760. P866.
- Biasco L, Izotova N, Rivat C, et al. Clonal expansion of T memory stem cells determines early anti-leukemic responses and long-term CAR T cell persistence in patients. *Nat Cancer*. 2021;2(6):629-642.
- Ledergor G, Fan Z, Wu K, et al. CD4+ CAR-T cell exhaustion associated with early relapse of multiple myeloma after BCMA CAR T-cell therapy. *Blood Adv*. 2024;8(13):3562-3575.
- Anderson ND, Birch J, Accogli T, et al. Transcriptional signatures associated with persisting CD19 CAR-T cells in children with leukemia. *Nat Med*. 2023;29(7):1700-1709.
- Boroughs AC, Larson RC, Marjanovic ND, et al. A distinct transcriptional program in human CAR T cells bearing the 4-1BB signaling domain revealed by scRNA-Seq. *Mol Ther*. 2020;28(12):2577-2592.
- Sheih A, Voillet V, Hanafi L-A, et al. Clonal kinetics and single-cell transcriptional profiling of CAR-T cells in patients undergoing CD19 CAR-T immunotherapy. *Nat Commun*. 2020;11(1):219.
- Freeman CL, Noble J, Menges M, et al. Tumor burden quantified by soluble B-cell maturation antigen and metabolic tumor volume determines myeloma CAR-T outcomes. *Blood*. 2025;145(15):1645-1657.
- Li X, Henderson J, Gordon MJ, et al. A single-cell atlas of CD19 chimeric antigen receptor T cells. *Cancer Cell*. 2023;41(11):1835-1837.
- Chu Y, Dai E, Li Y, et al. Pan-cancer T cell atlas links a cellular stress response state to immunotherapy resistance. *Nat Med*. 2023;29(6):1550-1562.
- Lalle G, Lautraite R, Bouherrou K, et al. NF- κ B subunits RelA and c-Rel selectively control CD4+ T cell function in multiple sclerosis and cancer. *J Exp Med*. 2024;221(6):e20231348.
- Rejeski K, Hansen DK, Bansal R, et al. The CAR-HEMATOTOX score as a prognostic model of toxicity and response in patients receiving BCMA-directed CAR-T for relapsed/refractory multiple myeloma. *J Hematol Oncol*. 2023;16(1):88.
- Yu X, Jain MD, Menges MA, et al. Comparison of axicabtagene ciloleucel and tisagenlecleucel patient CAR-T cell products by single-cell RNA sequencing. *J Immunother Cancer*. 2025;13(7):e011807.
- Rytlewski J, Hansen DK, Peres LC, et al. Preapheresis clinical patient profiles predict patient's manufacturing fitness and efficacy with idecabtagene vicleucel. *Blood Immunol Cell Ther*. 2026;2(2):100036.
- Dhodapkar KM, Cohen AD, Kaushal A, et al. Changes in bone marrow tumor and immune cells correlate with durability of remissions following BCMA CAR T therapy in myeloma. *Blood Cancer Discov*. 2022;3(6):490-501.
- Waldschmidt JM, Sotudeh N, Arora S, et al. Nivolumab to restore T-cell fitness in CAR-T refractory multiple myeloma. *Blood Adv*. 2025;9(5):1132-1136.
- Singh N, Frey NV, Engels B, et al. Antigen-independent activation enhances the efficacy of 4-1BB-costimulated CD22 CAR T cells. *Nat Med*. 2021;27(5):842-850.
- Frigault MJ, Lee J, Basil MC, et al. Identification of chimeric antigen receptors that mediate constitutive or inducible proliferation of T cells. *Cancer Immunol Res*. 2015;3(4):356-367.
- Long AH, Haso WM, Shern JF, et al. 4-1BB costimulation ameliorates T cell exhaustion induced by tonic signaling of chimeric antigen receptors. *Nat Med*. 2015;21(6):581-590.
- Gomes-Silva D, Mukherjee M, Srinivasan M, et al. Tonic 4-1BB costimulation in chimeric antigen receptors impedes T cell survival and is vector-dependent. *Cell Rep*. 2017;21(1):17-26.

© 2026 American Society of Hematology. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.